

Directions: Read each description carefully. Use light construction lines first, then darken the final visible edges.
Curved features should appear as ellipses in isometric view. Work proportionally; exact scale is not required.

1. Rectangular Prism

What to draw:

Draw a rectangular prism that is 4 units long, 3 units deep, and 3 units high. Begin with the front lower corner, then construct the top, right, and left receding edges using the isometric axes. Darken all final visible edges.

ISOMETRIC SKETCH SPACE



2. Stepped Block

What to draw:

Draw a base block that is 4 units long, 3 units deep, and 1 unit high. On the left half of the base, add a raised step that uses the full depth and rises 2 more units above the base. The finished object should have one lower section and one taller stepped section.

ISOMETRIC SKETCH SPACE



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3. Corner Notch Block

What to draw:

Draw a 3 x 3 x 3 block. Remove the upper front-right corner by cutting out a notch that is about 1 unit long, 1.5 units deep, and 1.5 units high. Keep the remaining edges clean and show the cut clearly in isometric view.

ISOMETRIC SKETCH SPACE



4. Top-Surface Circle / Ellipse

What to draw:

Draw a flat top surface as an isometric square or rhombus. Inside it, sketch a circle in isometric view, which should appear as an ellipse. The ellipse should be centered and tangent near the midpoint of each side of the bounding rhombus.

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5. Vertical Cylinder

What to draw:

Draw a vertical cylinder that is about 4 units tall and 2 units in diameter. Start by drawing the bottom ellipse and a matching top ellipse directly above it. Connect the two ellipses with vertical tangent edges to complete the cylinder.

ISOMETRIC SKETCH SPACE



6. Bracket with Circular Hole

What to draw:

Draw a bracket-like object with a low rectangular base and an upright plate rising from the back portion of the base. In the front face of the upright plate, add a centered circular through-hole shown in isometric view as an ellipse. Include centerlines through the hole.

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7. Horizontal Cylinder

What to draw:

Draw a cylinder lying horizontally along one of the 30-degree isometric axes. Make the cylinder about 4 units long and 2 units in diameter. Draw matching elliptical end faces, align their centers along the cylinder axis, and connect them with tangent edges.

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8. Flanged Support with Curved Top

What to draw:

Draw a rectangular base plate about 5 units long, 3 units deep, and 1 unit high. Add an upright support centered near the back edge. Give the support a rounded or semicircular top and place a circular through-hole near the center of that curved upper portion. Use an ellipse for the hole and keep the support centered on the base.

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